



MATLAB Drive > dsp8_1.m*

```
1 clc;
2 clear;
3 close all;
4
5 % Original Sequence
6 n = 0:40;
7 x = sin(0.3*pi*n);
8
9 % Decimation by 2
10 y = x(1:2:end);
11 nd = 0:length(y)-1;
12
13 % Interpolation by 2
14 v = zeros(1,2*length(x));
15 v(1:2:end) = x;
16 ni = 0:length(v)-1;
17
18 % Figure 1: Original Sequence
19 figure;
20 stem(n,x,'filled');
21 grid on;
22 title('Original Sequence x[n]');
23 xlabel('n');
24 ylabel('Amplitude');
25
26 % Figure 2: Decimated Sequence
27 figure;
28 stem(nd,y,'filled');
29 grid on;
30 title('Decimated Sequence by 2');
31 xlabel('n');
32 ylabel('Amplitude');
33
34 % Figure 3: Interpolated Sequence
35 figure;
36 stem(ni,v,'filled');
37 grid on;
38 title('Interpolated Sequence by 2');
39 xlabel('n');
40 ylabel('Amplitude');
41
42 % Figure 4: Comparison
43 figure;
44 stem(n,x,'b');
45 hold on;
46 stem(0:2:length(v)-2,v(1:2:end),'r');
47 legend('Original x[n]', 'Interpolated Samples')
```



Figures

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Figure 1: dsp8_1

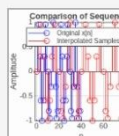
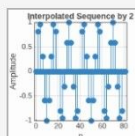
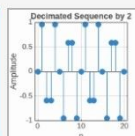
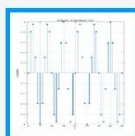
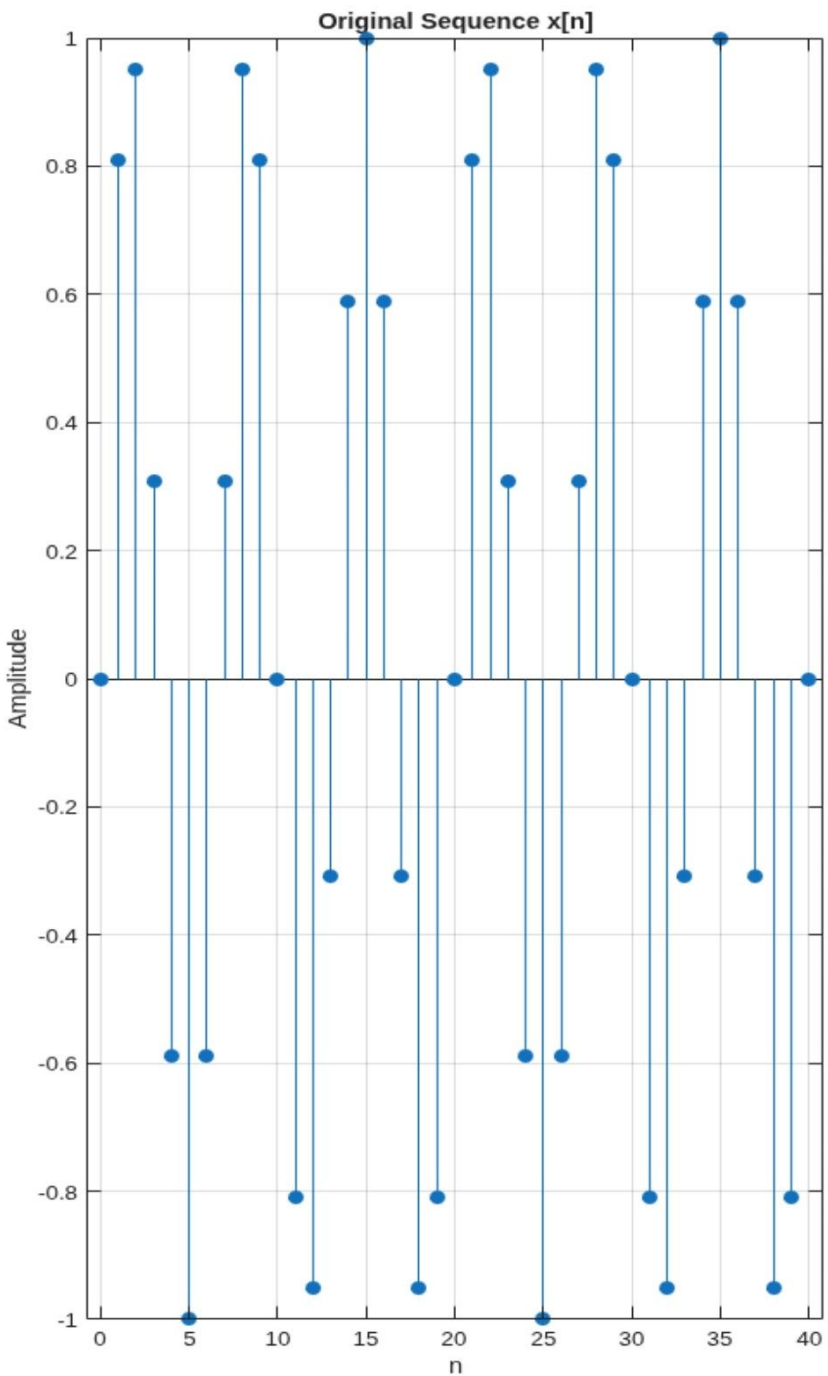
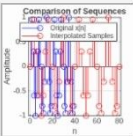
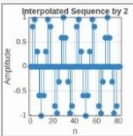
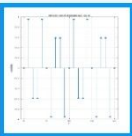
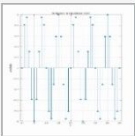
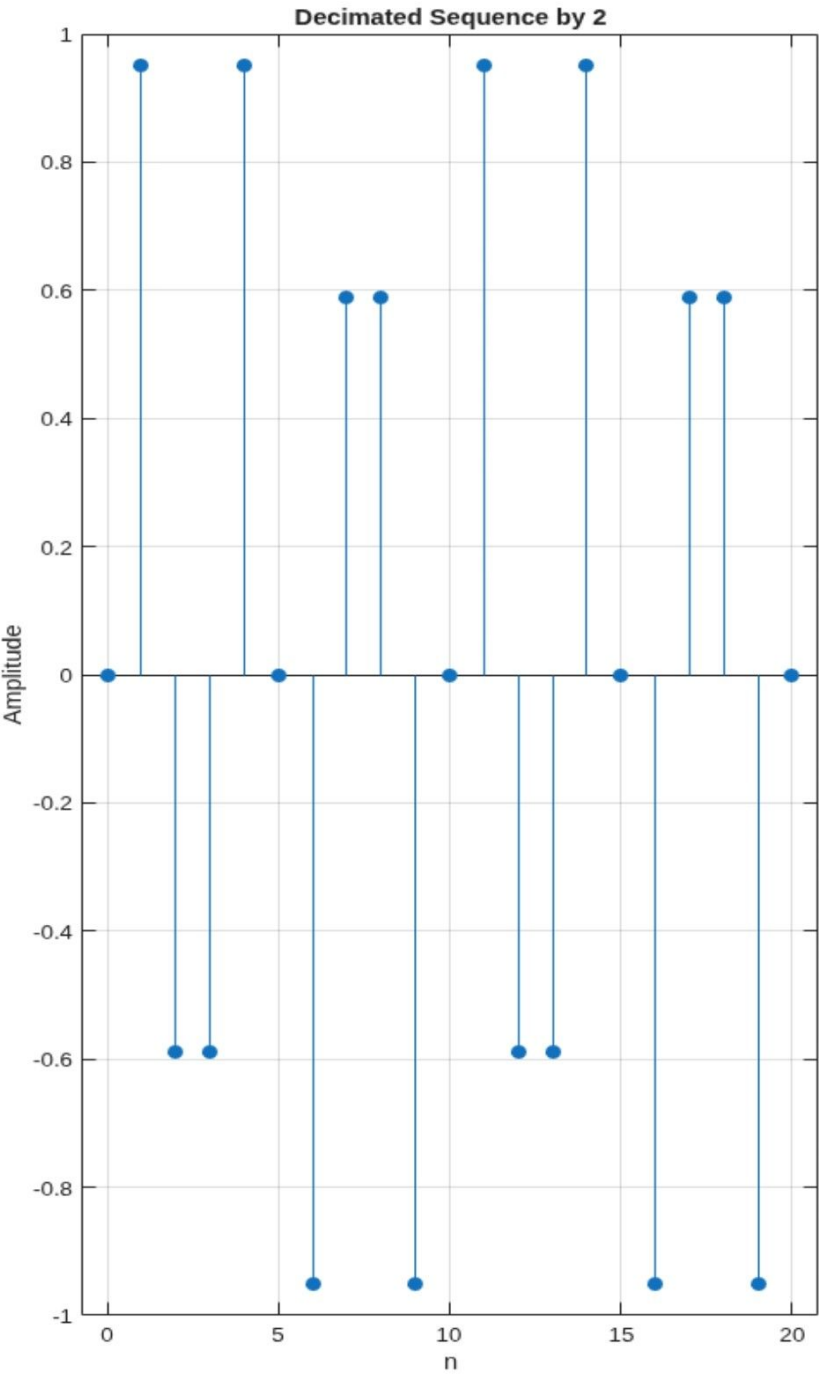
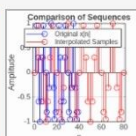
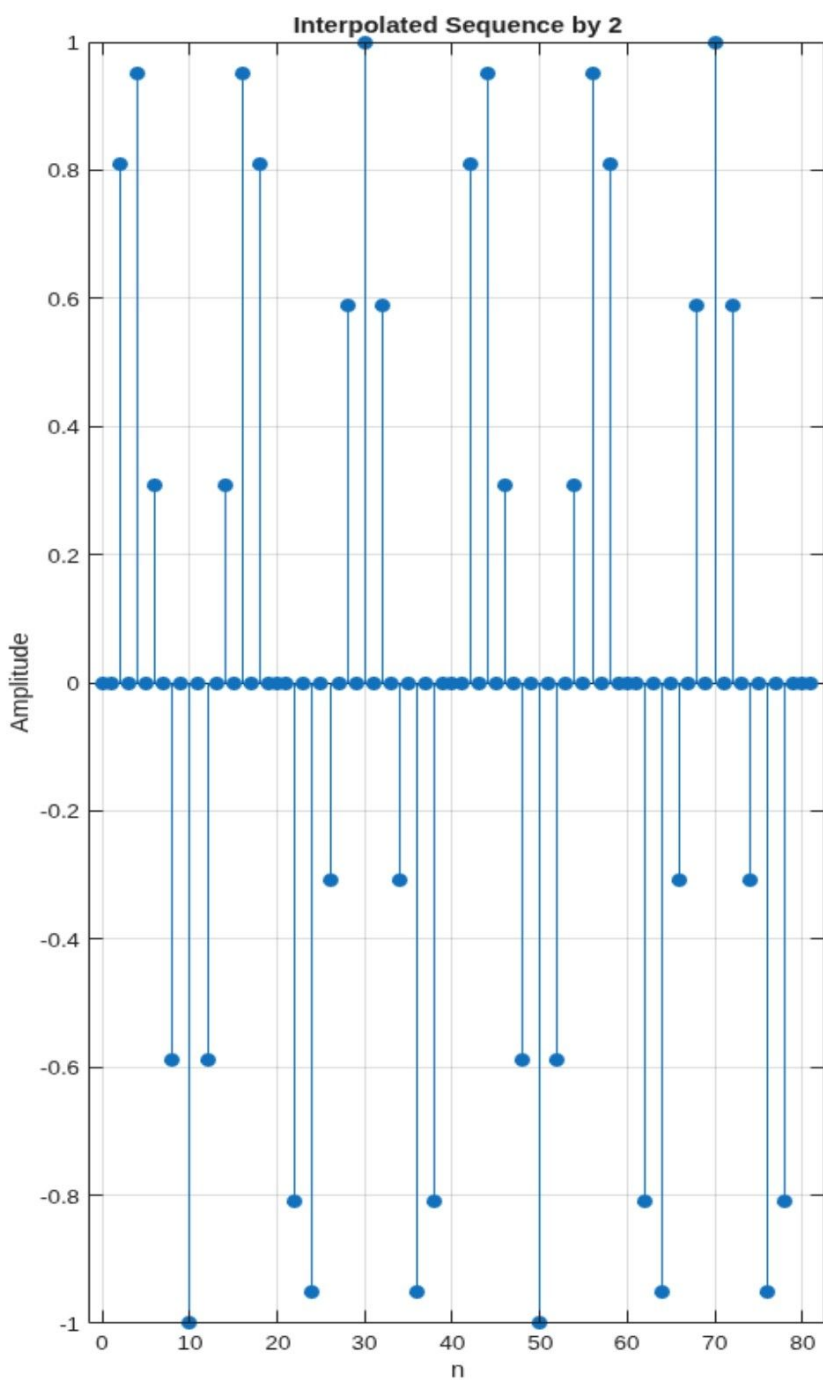


Figure 2: dsp8_1





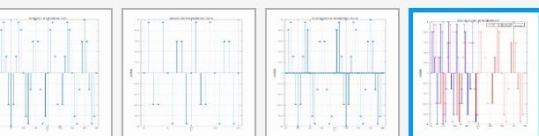
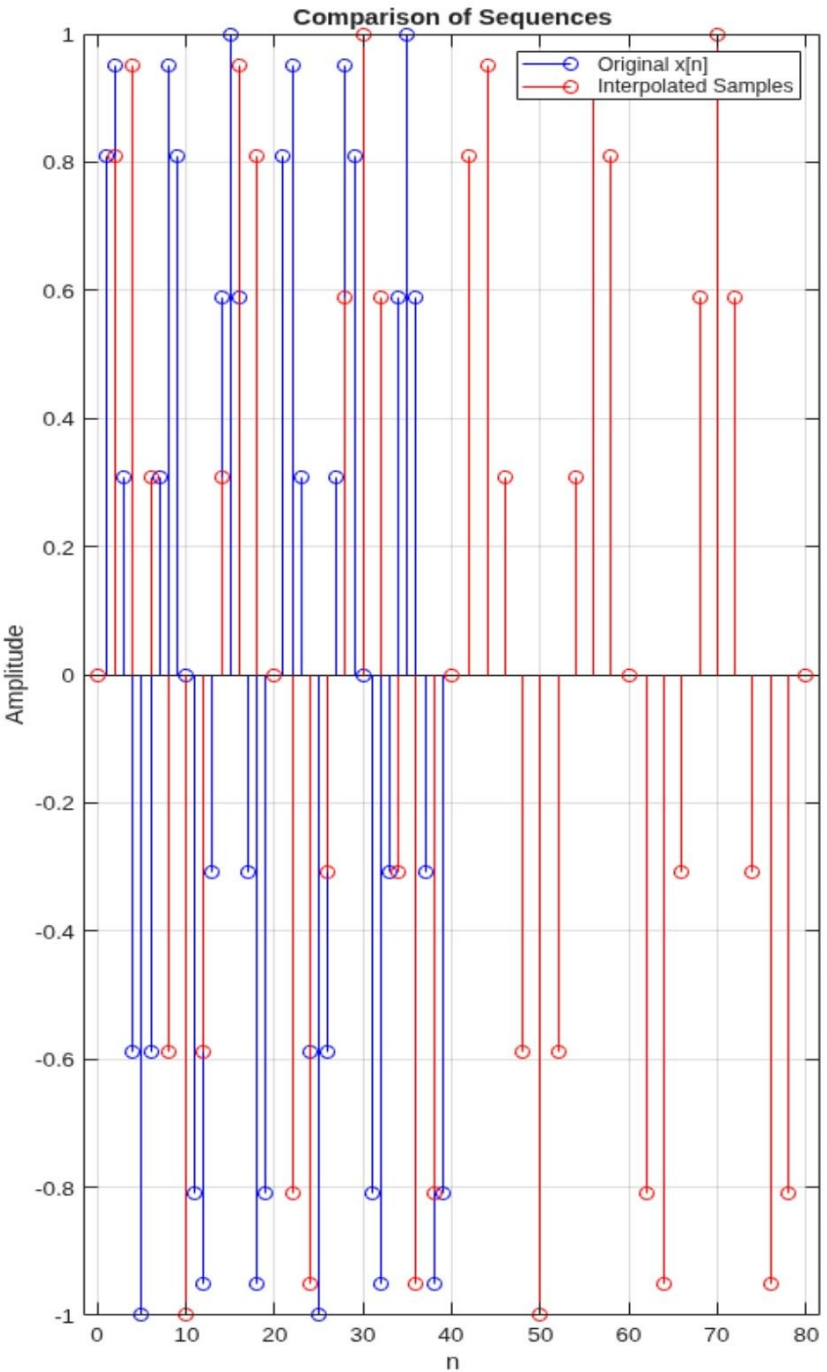


Figures

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Figure 4: dsp8_1





MATLAB Drive > dsp8_2.m*

```
1 clc;
2 clear;
3 close all;
4
5 %% Input Signal
6 Fs = 1000; % Sampling Frequency
7 t = 0:1/Fs:1; % Time Vector
8
9 % Test Signal
10 x = sin(2*pi*50*t) + 0.5*sin(2*pi*120*t);
11
12 %% FIR Low-Pass Filter Design
13 N = 20; % Filter Order
14 Fc = 0.4; % Normalized Cutoff
15 h = fir1(N,Fc,hamming(N+1));
16
17 %% Decimation Factor
18 M = 2;
19 y_dec = upfirdn(x,h,1,M);
20
21 %% Interpolation Factor
22 L = 2;
23 y_int = upfirdn(y_dec,h,L,1);
24
25 %% Adjust Length
26 if length(y_int) > length(x)
27     y_int = y_int(1:length(x));
28 elseif length(y_int) < length(x)
29     x = x(1:length(y_int));
30 end
31
32 %% Time Axes
33 t1 = (0:length(x)-1)/Fs;
34 t2 = (0:length(y_dec)-1)/(Fs/M);
35 t3 = (0:length(y_int)-1)/Fs;
36
37 %% Graphs
38 figure('Name','Polyphase Decimation and Interpolation')
39
40 subplot(4,1,1)
41 plot(t1,x,'b','LineWidth',1.5)
42 title('Original Input Signal')
43 xlabel('Time (s)')
44 ylabel('Amplitude')
45 grid on
46
47 subplot(4,1,2)
```




Figures

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Polyphase Decimation and Interpolation: dsp8_2

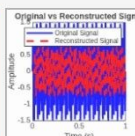
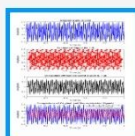
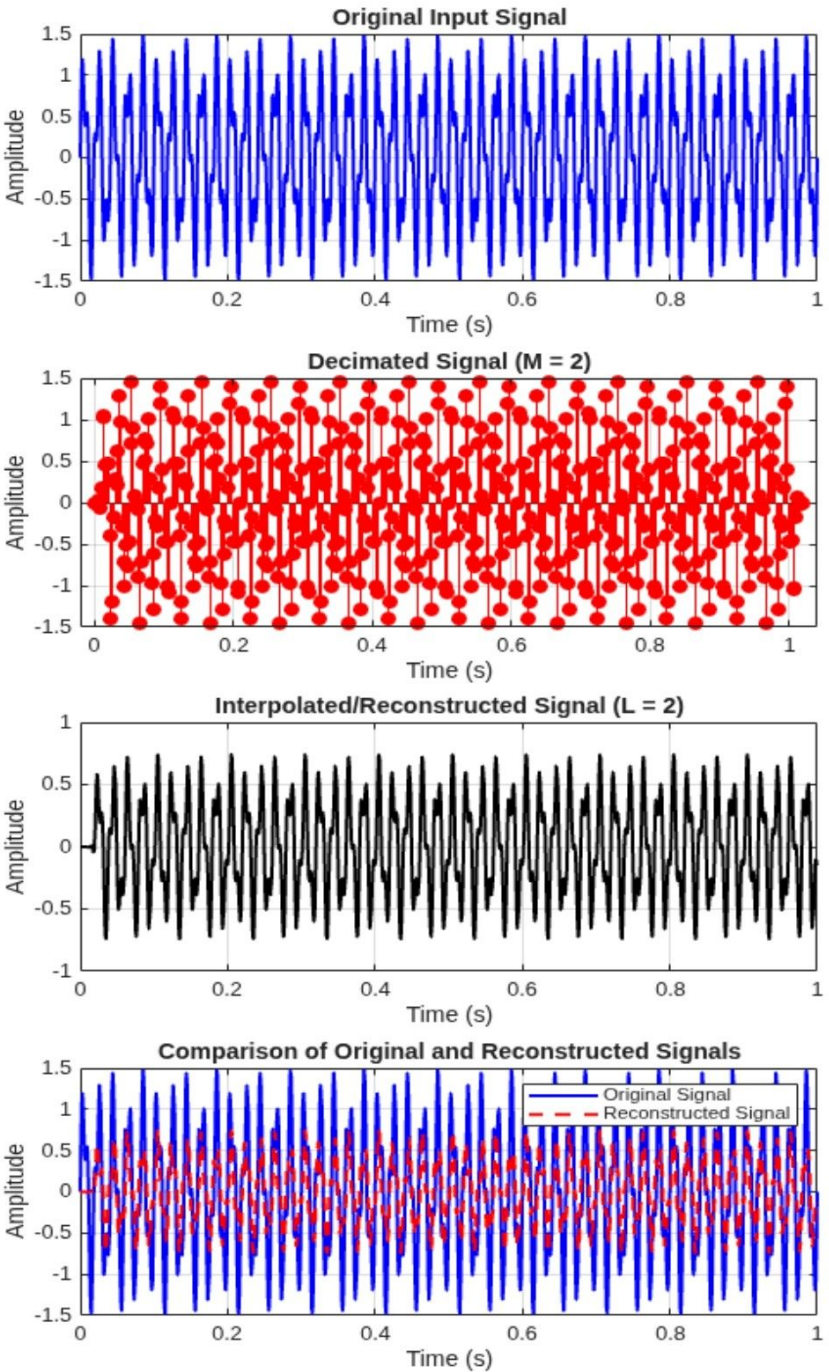


Figure 2: dsp8_2

